

Algorithms

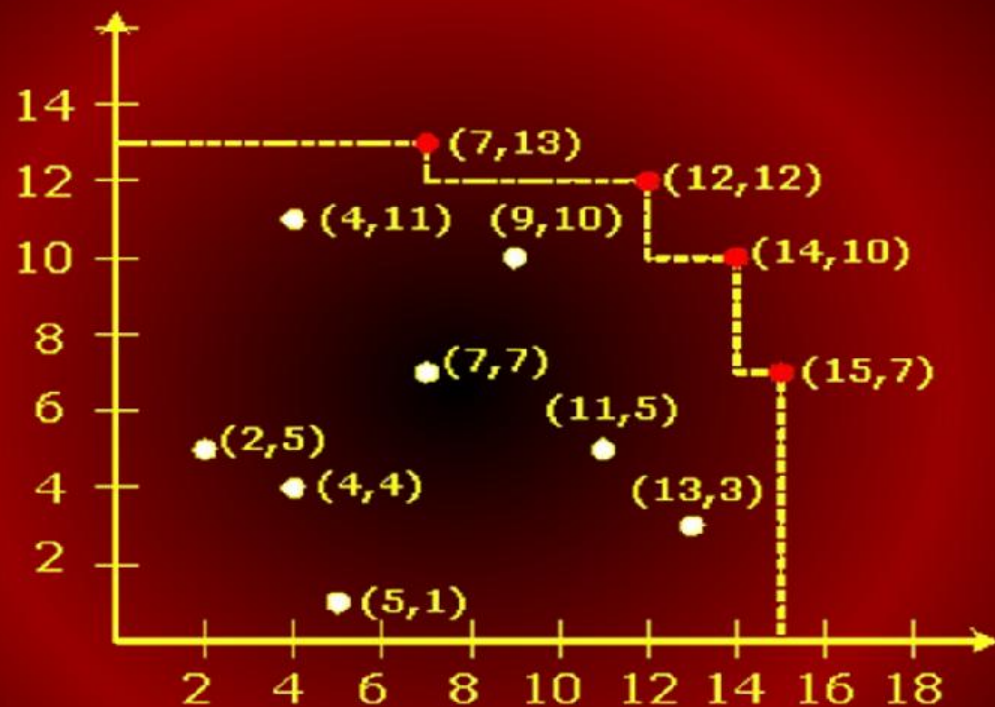
Lecture # 02

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Example – 2 dimension maxima

Random Access Machine (RAM)

Graph



Basic RAM Operations

Basic RAM Operations

Basic operations include

- assigning a value to a variable,
- computing any basic arithmetic operation (+, -, × and division) on integer values of any size
- performing any comparison or boolean operations
- accessing an element of an array (e.g. A[10])

RAM Model

RAM Model

- We assume that each basic operation takes the same constant time to execute.
- The RAM model does a good job of describing the computational power of most modern(nonparallel) machines.

RAM Model

RAM Model

Basic operations in C++ that are assumed to take up same amount of CPU time.

- $x = y;$
- $z = a + b; z = a - b;$
- $z = a * b; z = a / b;$
- $z = w[10];$
- $(x \leq y)$

Example: 2-dimension maxima

Example: 2-dimension maxima

Let us do an example that illustrates how we analyze algorithms.

- Suppose you want to buy a car.
- You want to pick the fastest car.
- But fast cars are expensive; you want the cheapest.
- You cannot decide which is more important: speed or price.

Example: 2-dimension maxima

Example: 2-dimension maxima

- Definitely *do not* want a car if there is another that is both faster and cheaper.
- We say that the fast, cheap car *dominates* the slow, expensive car relative to your selection criteria.
- So, given a collection of cars, we want to list those cars that are not dominated by any other.

Example: 2-dimension maxima

Example: 2-dimension maxima

Here is how we might model this as a formal problem.

- Let a point p in 2-dimensional space be given by its integer coordinates, $p = (p.x, p.y)$.
- A point p is said to be ***dominated*** by point q if $p.x \leq q.x$ and $p.y \leq q.y$.

Example: 2-dimension maxima

Example: 2-dimension maxima

- Given a set of n points, $P = \{p_1, p_2, \dots, p_n\}$ in 2-space a point is said to be ***maximal*** if it is not dominated by any other point in P .

Example: 2-dimension maxima

Example: 2-dimension maxima

The car selection problem can be modelled this way:

For each car we associate (x, y) pair where

- x is the speed of the car and
- y is the negation of the price.

Example: 2-dimension maxima

Example: 2-dimension maxima

- High y value means a cheap car and low y means expensive car.
- Think of y as the money left in your pocket after you have paid for the car.
- Maximal points correspond to the fastest and cheapest cars.

Example: 2-dimension maxima

Example: 2-dimension maxima

2-dimensional Maxima:

- Given a set of points $P = \{p_1, p_2, \dots, p_n\}$ in 2-space,
- output the set of maximal points of P ,
- i.e., those points p_i such that p_i is not dominated by any other point of P .

Example: 2-dimension maxima

Example: 2-dimension maxima

- Our description of the problem is at a fairly mathematical level.
- We have intentionally not discussed how the points are represented.
- We have not discussed any input or output formats.
- We have avoided programming and other software issues.

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